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In [1]: import numpy as np
import matplotlib.pyplot as plt

#(a) Function of modulated signal
fs = 2000
t = np.linspace(0, 1, fs, endpoint=False)

ym = 10*np.sin(200*np.pi*t) + 40*np.sin(100*np.pi*t)
yc = 20*np.sin(400*np.pi*t)

y = (20 + ym) * np.sin(400*np.pi*t)

#(b) Plot Vc(t), Vm(t) and y(t) for t in [0, 0.1]
t_plot = t[t <= 0.1]
plt.figure(figsize=(12, 7))

plt.subplot(3, 1, 1)
plt.plot(t_plot, yc[t <= 0.1])
plt.title('Carrier Wave Vc(t)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude (V)')

plt.subplot(3, 1, 2)
plt.plot(t_plot, ym[t <= 0.1])
plt.title('Modulating Signal Vm(t)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude (V)')

plt.subplot(3, 1, 3)
plt.plot(t_plot, y[t <= 0.1])
plt.title('Modulated Signal y(t)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude (V)')
plt.show()

#(c) FFT algorithm
#(i) result plot
fft_result = np.fft.fft(y)
frequencies = np.fft.fftfreq(len(fft_result), 1/fs)
positive_mask = frequencies > 0
positive_frequencies = frequencies[positive_mask]
positive_magnitudes = np.abs(fft_result)[positive_mask] * 2 / fs

plt.plot(positive_frequencies, positive_magnitudes)
plt.title('Frequency Components of Modulated Signal')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Amplitude (V)')
plt.grid(True)

#(ii) Lowest frequency
threshold = 0.1
significant_indices = positive_magnitudes > threshold
for freq, amp in zip(positive_frequencies[significant_indices], positive_magnitudes[significant_indices]):
    plt.text(freq, amp, f'{freq:.0f} Hz', ha='center', va='bottom')

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plt.tight_layout()
plt.show()

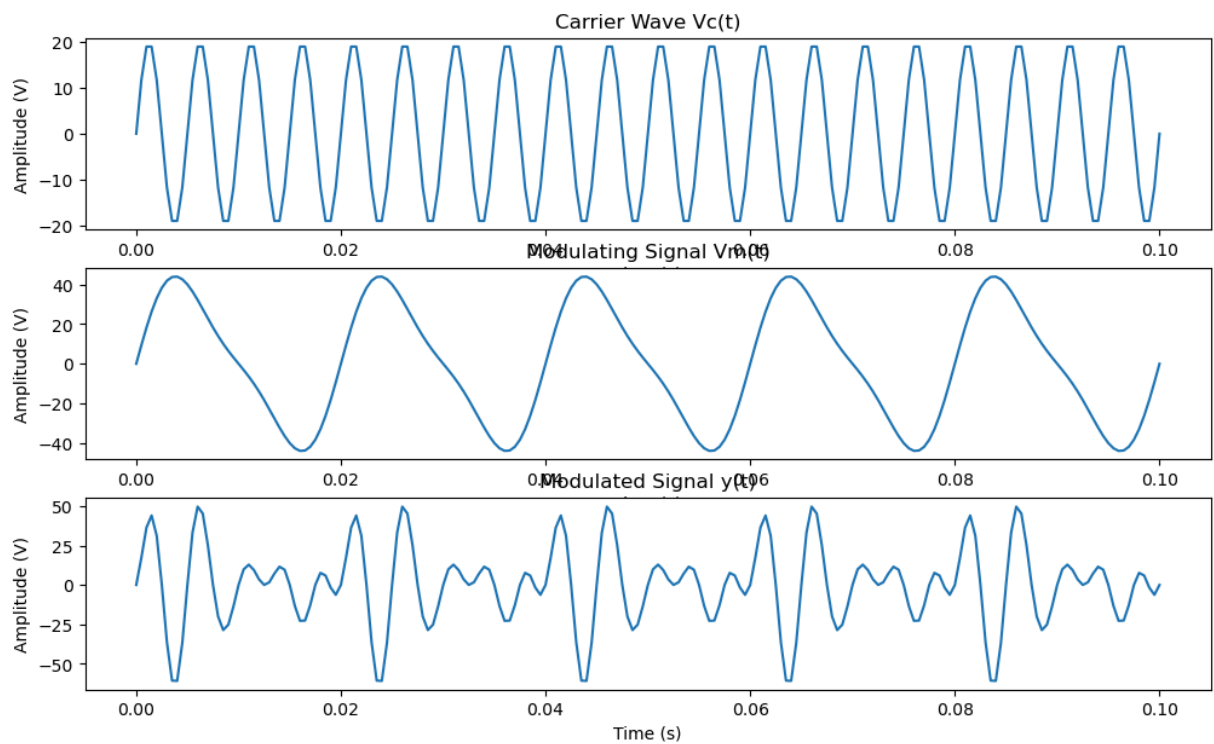
significant_freqs = positive_frequencies[significant_indices]
significant_mags = positive_magnitudes[significant_indices]
print("Detected Frequencies and Amplitudes:")
for freq, amp in zip(significant_freqs, significant_mags):
    print(f"Frequency: {freq:.1f} Hz, Amplitude: {amp:.2f} V")

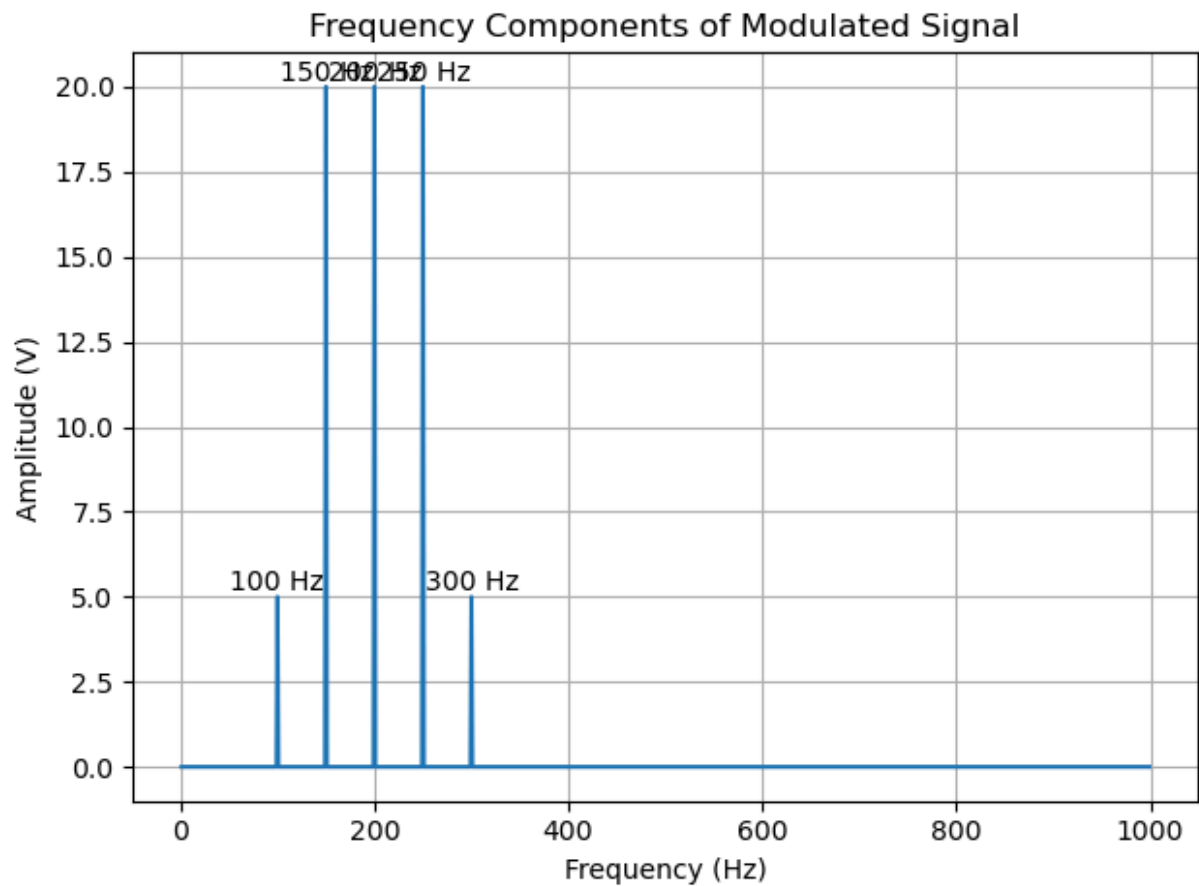
print(f"ii. Lowest frequency: {min(significant_freqs):.1f} Hz")

#(iii) Highest frequency
print(f"iii. Highest frequency: {max(significant_freqs):.1f} Hz")

#(iv) Band width based on the Fourier spectrum
print(f"iv. Bandwidth: {(max(significant_freqs) - min(significant_freqs)):.1f} Hz")

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Detected Frequencies and Amplitudes:

Frequency: 100.0 Hz, Amplitude: 5.00 V

Frequency: 150.0 Hz, Amplitude: 20.00 V

Frequency: 200.0 Hz, Amplitude: 20.00 V

Frequency: 250.0 Hz, Amplitude: 20.00 V

Frequency: 300.0 Hz, Amplitude: 5.00 V

ii. Lowest frequency: 100.0 Hz

iii. Highest frequency: 300.0 Hz

iv. Bandwidth: 200.0 Hz

In []: